Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN Practical

Practical – V

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Class : TYIT	Date/Time of Submission : 20/08/2026

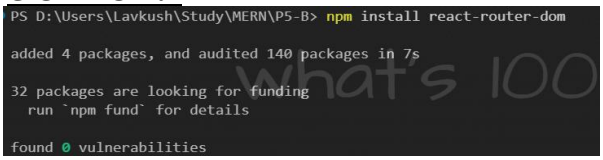
Aim: React Routing and Data Handling.

5.A). Install and configure React Router in the React application..

SOURCE CODE:-

cd projname
npm install react-router-dom

OUTPUT:-



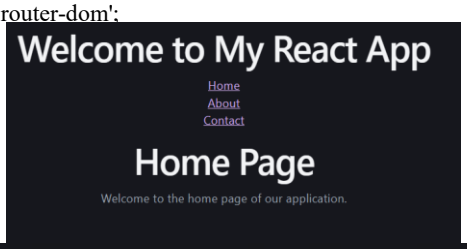
5.B). Create multiple pages and implement navigation between them.

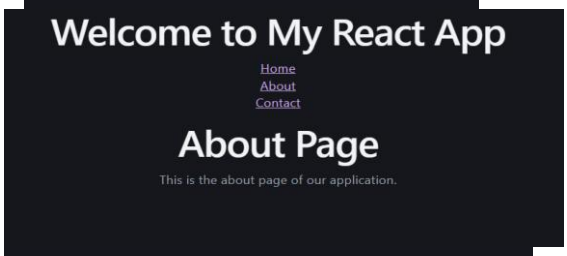
SOURCE CODE:-

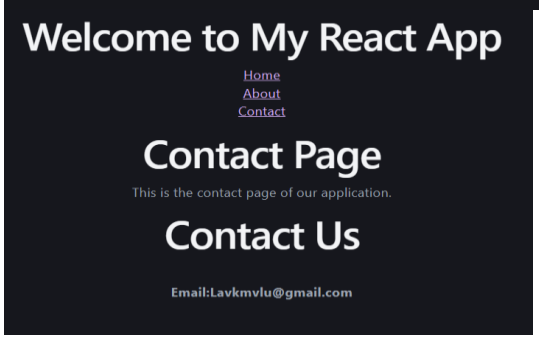
App.JSX

```
import { BrowserRouter , Routes, Route , Link } from 'react-router-dom';
import Home from './pages/Home';
import About from './pages/About';
import Contact from './pages/Contact';
function App() {
 return (
  <BrowserRouter>
   <nav>
    <h1>Welcome to My React App</h1>
    <Link to="/">Home</Link>
    <br />
    <Link to="/About">About</Link>
    <br />
    <Link to="/Contact">Contact</Link>
   </nav>
   <main>
    <Routes>
     <Route path="/" element={<Home />} />
     <Route path="/About" element={<About />} />
     <Route path="/Contact" element={<Contact />} />
    </Routes>
   </main>
  </BrowserRouter>
 );
}
export default App;
```

OUTPUT:-







About.JSX

```
 return (
  <div>
```

```
function About() {
    <h1>About Page</h1>
    <p>This is the about page of our application.</p>
```

```
    </div>
  );
}
export default About;
```

Contact.JSX

```
function Contact() {
  return (
    <div>
      <h1>Contact Page</h1>
      <p>This is the contact page of our application.
        <h1>Contact Us</h1>
        <br />
        <b>Email:Lavkmvlu@gmail.com</b>
      </p>
    </div>
  );
}
export default Contact;
```

Home.JSX

```
function Home() {
  return (
    <div>
      <h1>Home Page</h1>
      <p>Welcome to the home page of our application.</p>
    </div>
  );
}
export default Home;
```

5.C). Use Link and Routes to move between pages.

SOURCE CODE:-

App.jsx

```
import { BrowserRouter, Routes, Route, Link } from 'react-
router-dom';
import Dashboard from './pages/Dashboard';
import Jobs from './pages/Jobs';
import Profile from './pages/Profile';
import Stats from './pages/Stats';
function App() {
  return (
    <BrowserRouter>
      <nav>
        <h1>Welcome to My React App</h1>
        <Link to="/">Dashboard</Link>
        <br />
        <Link to="/jobs">Jobs</Link>
        <br />
        <Link to="/profile">Profile</Link>
        <br />
        <Link to="/stats">Stats</Link>
      </nav>
      <main>
        <Routes>
          <Route path="/" element={<Dashboard />} />
          <Route path="/jobs" element={<Jobs />} />
          <Route path="/profile" element={<Profile />} />
          <Route path="/stats" element={<Stats />} />
        </Routes>
      </main>
    </BrowserRouter>
  );
}
export default App;
```

Dashboard.jsx

```
function Dashboard() {
  return (
    <div>
      <h1>Placement Dashboard</h1>
      <p>Welcome to the Student Placement Portal!</p>
      <hr />
      <h2>Placement Summary</h2>
      <p>Total Students Placed: 120</p>
      <p>Total Companies Visited: 15</p>
      <p>Highest Package Offered: ₹75,000</p>
    </div>
  );
}
export default Dashboard;
```

Jobs.jsx

```
function Jobs() {
  return (
    <div>
      <h1>Job Page</h1>
      <p>Welcome to the Job page.</p>
      <p>Available Jobs: 25</p>
    </div>
  );
}
export default Jobs;
```

Stats.jsx

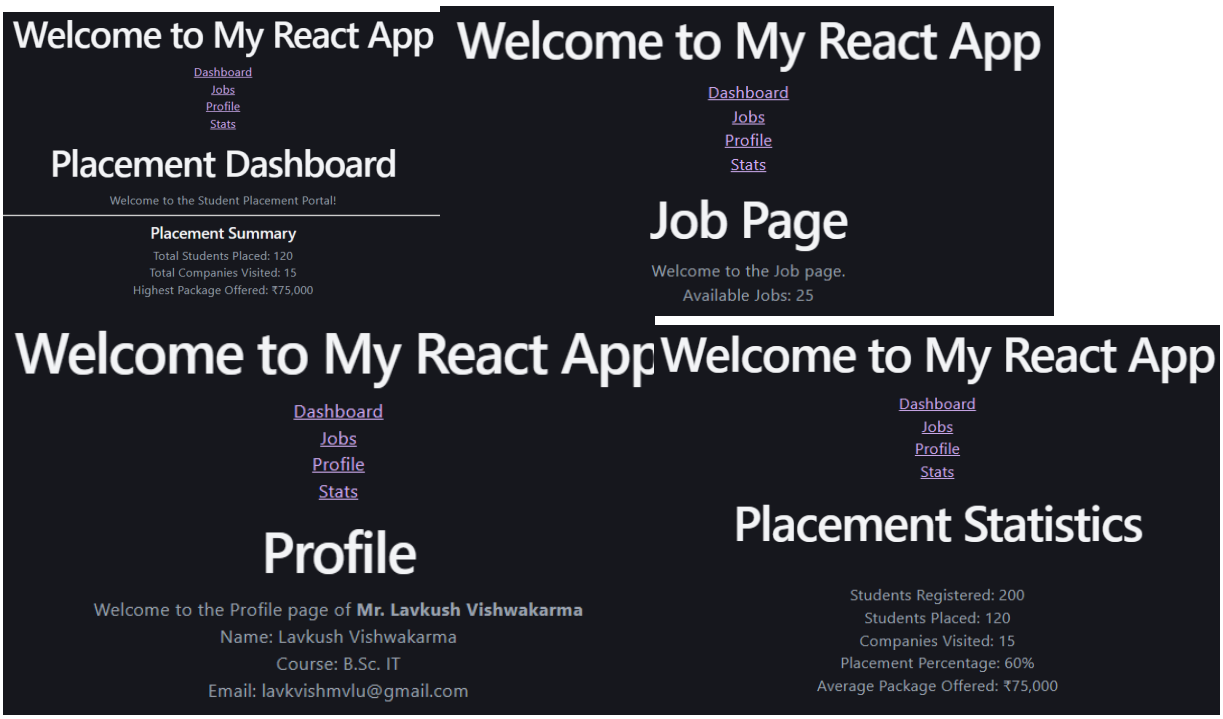
```
function Stats() {
  return (
    <div>
      <h1>Placement Statistics</h1>
      <br />
      <p>Students Registered: 200</p>
      <p>Students Placed: 120</p>
      <p>Companies Visited: 15</p>
      <p>Placement Percentage: 60%</p>
      <p>Average Package Offered: ₹75,000</p>
    </div>
  );
}
export default Stats;
```

Profile.jsx

```
function Profile() {
  return (
    <div>
      <h1>Profile</h1>
      <p>
        Welcome to the Profile page of <b>Mr. Lavkush
Vishwakarma </b>
      </p>
      <p>Name: Lavkush Vishwakarma</p>
```

```
    <p>Course: B.Sc. IT</p>
    <p>Email: lavkvishmvlu@gmail.com</p>
   </div>
  );
}
export default Profile;
```

OUTPUT:-



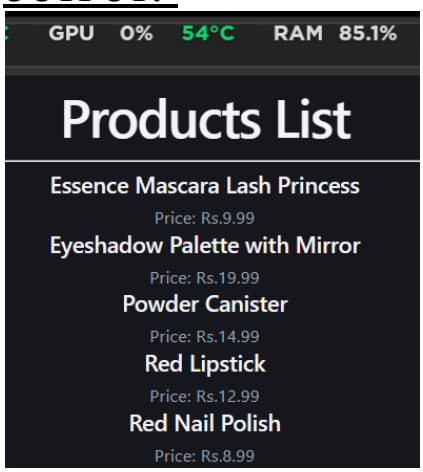
5.D). Fetch sample data from a public API using Axios..

SOURCE CODE:-

App.jsx

```
import axios from "axios";
import { useEffect, useState } from "react";
function App() {
  const [products, setProducts] = useState([]);
  useEffect(() => {
    axios.get("https://dummyjson.com/products"
).then((response) => {
      setProducts(response.data.products);
    });
  }, []);
  return (
    <div>
      <h1>Products List</h1>
      <hr />
      {products.map((product)=> (
        <div key={product.id}>
          <h2>{product.title}</h2>
          <p>Price: Rs.{product.price}</p>
        </div>
```

OUTPUT:-




```
      ))}
    </div>
  );
}
export default App;
```

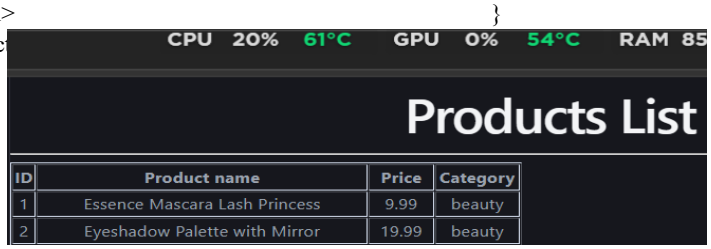
5.E). Display fetched data in table or list format.

SOURCE CODE:-

App.jsx

```
import axios from "axios";
import { useEffect, useState } from "react";
function App() {
  const [products, setProducts] = useState([]);
  useEffect(() => {
axios.get("https://dummyjson.com/products").then((respons
e) => {
      setProducts(response.data.products);
    });
  }, []);
  return (
    <div>
      <h1>Products List</h1>
      <hr />
      <table border="1">
        <thead>
          <tr>
            <th>ID</th>
            <th>Produc
            <th>Price</th>
            <th>Category</th>
          </tr>
        </thead>
        <tbody>
          {products.map((product) =>(
            <tr key={product.id}>
              <td>{product.id}</td>
              <td>{product.title}</td>
              <td>{product.price}</td>
              <td>{product.category}</td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
  );
}
```

OUTPUT:-



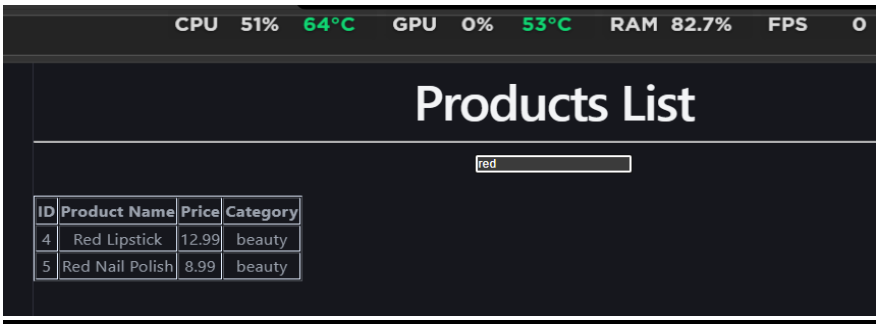
5.F). Implement search or filter functionality on displayed data.

SOURCE CODE:-

```
import axios from "axios";
import { useEffect, useState } from "react";
function App() {
  const [products, setProducts] = useState([]);
  const [search, setSearch] = useState("");
  useEffect(() => {
    axios
      .get("https://dummyjson.com/products")
      .then((response) => {
        setProducts(response.data.products);
      });
  }, []);
  // Filter products based on search text
  const filteredProducts = products.filter(
    (product) =>
product.title.toLowerCase().includes(search.toLowerCase()
) ||
product.category.toLowerCase().includes(search.toLowerC
ase())
  );
  return (
    <div>
      <h1>Products List</h1>
      <hr />
      {/* Search Box */}
      <input
        type="text"
        placeholder="Search products..."
        value={search}
        onChange={(e) => setSearch(e.target.value)}
      />
      <br />
      <br />
      <table border="1">
        <thead>
          <tr>
            <th>ID</th>
            <th>Product Name</th>
            <th>Price</th>
            <th>Category</th>
          </tr>
        </thead>
        <tbody>
          {filteredProducts.map((product) => (
            <tr key={product.id}>
              <td>{product.id}</td>
              <td>{product.title}</td>
              <td>{product.price}</td>
              <td>{product.category}</td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
```

```
  );                                                    export default App;
}
```

OUTPUT:-

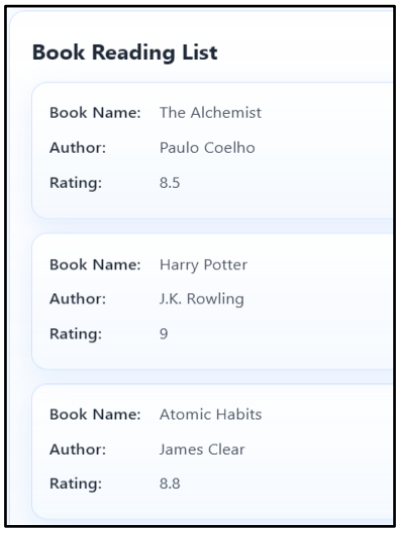


Extra=>4.E). TO-DO LIST(BOOK READING LIST).

SOURCE CODE:-

```
import React, { useState } from "react";
import Header from "./components/Header";
import Footer from "./components/Footer";
import Counter from "./components/Counter";
import UserForm from "./components/UserForm";
import UserList from "./components/UserList";
import "./App.css";
const App = () => {
  const [books] = useState([
    {
      id: 1,
      name: "The Alchemist",
      author: "Paulo Coelho",
      rating: 8.5,    },
    {
      id: 2,
      name: "Harry Potter",
      author: "J.K. Rowling",
    },
    {
      id: 3,
      name: "Atomic Habits",
      author: "James Clear",
      rating: 8.8,    },
    {
      id: 4,
      name: "The Psychology of Money",
      author: "Morgan Housel",
      rating: 8.7,    },
  ]);
  const [users, setUsers] = useState([
    { id: 1, name: "Alice Smith", email:
"alice@example.com" },
    { id: 2, name: "Bob Jones", email:
"bob@example.com"},
  ]);
  const addUser = (user) => {
    setUsers((prev) => [...prev, user]);  };
  return (
    <div className="App">
      <div className="container">
        <Header />
        <div className="card">
```

OUTPUT:-




```
      rating: 9.0,
        <h2>Book Reading List</h2>
        {books.map((book) => (
          <div className="book-card" key={book.id}>
            <p>
              <span className="label">Book Name:</span>
              <span className="value">{book.name}</span>
            </p>
            <p>
              <span className="label">Author:</span>
              <span
className="value">{book.author}</span>
            </p>
            <p>
              <span className="label">Rating:</span>
              <span className="value">{book.rating}</span>
            </p>
          </div>   ))}
        </div>
        <Counter />
        <UserForm onAddUser={addUser} />
        <UserList users={users} />
        <Footer />
      </div>
    </div>  );};
export default App;
```